Venkatasai Mannava

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
Vignan Cooperative junior college	Class XII (Board of Intermediate Education, Andhra Pradesh)	2019	10.00%
Sri Somanatha Green field public school	Class X (Central Board of secondary education)	2017	10.00%

ABOUT

Communications and Signal Processing Engineer with hands-on experience in real-time GNSS receiver development.Currently pursuing an M.Tech at IIT Hyderabad, with practical experience in implementing GPS L1 C/A baseband signal-processing algorithms on DSP platforms.Proficient in C, Python, MATLAB, and SDR-based signal processing, with a focus on the development, optimisation, and real-time implementation of communication and signal-processing systems.

TECHNICAL SKILLS

- **Programming Languages :** C, Verilog,Python, MATLAB
- **Embedded DSP :** Implementation and optimization of real-time signal processing algorithms using fixed-point arithmetic, with emphasis on computational efficiency, numerical precision, and memory-constrained embedded DSP architectures.
- **Low-Level Debugging :** Debugging C programs on target hardware through register-level inspection.
- **Memory Mapping :** Developing linker scripts for custom memory mapping and efficient memory utilization.
- **RF Test Equipment :** Hands-on experience with Rohde & Schwarz spectrum analyzers and signal generators for signal characterisation and power measurements.
- **Software-Defined Radio :** Hands-on experience with SDR platforms for real-time signal transmission, reception, and I/Q data processing.
- **Scripting and Automation :** Proficient in Python and Bash scripting for test automation, data processing, and Monte Carlo simulation for baseband processing blocks.
- **Development Tools :** Linux, GCC, GDB, Make, Git, VS Code,Bash

EXPERIENCE

5G Testbed R & D Lab, IIT Hyderabad Aug 2024 - Present

M.Tech RA

- Ported and optimised the **GPS L1 C/A base band receiver** for a resource-constrained **CEVA DSP** using fixed-point arithmetic in C, targeting a platform with a **153 MHz clock frequency and 512 KB on chip memory (code and data memory)**.
- Optimised the receiver signal-processing pipeline to achieve a **tracking execution time of 160 us per satellite**, enabling efficient real-time multi-channel processing within the available computational resources.
- Performed extensive **Monte Carlo testing** across different geographic locations and signal conditions, demonstrating positioning performance comparable to the floating-point implementation.
- Conducted mathematical analysis of the complete receiver signal-processing chain and characterised the **noise performance and sensitivity** of individual processing stages.
- Achieved a receiver sensitivity of approximately **-140 dBm for acquisition** and **-150 dBm for tracking**, with a **median positioning accuracy of approximately 7 m**.
- Identified, analysed, and resolved complex **algorithmic and implementation-level challenges** during DSP integration, addressing fixed-point precision, computational complexity, memory utilisation, and real-time processing constraints.

5G Testbed R & D Lab, IIT Hyderabad Feb 2024 - July 2024

Project Associate

- Developed a strong understanding of the GPS L1 C/A signal structure and receiver signal-processing algorithms.
- Designed and implemented a complete GPS L1 C/A base band receiver in C using floating-point arithmetic, comprising acquisition, tracking, bit synchronisation, frame synchronisation, and PVT estimation modules.
- Validated the receiver using live GPS L1 C/A signals captured with a Blade RF SDR and evaluated receiver performance through extensive Monte Carlo simulations.
- Achieved a median positioning accuracy of approximately 7 m while tracking signals from six satellites.

PROJECTS

GPS L1 C/A Receiver on DSP

- As part of my M.Tech thesis, developed and optimised a **real-time GPS L1 C/A base band receiver on a CEVA DSP** using fixed-point arithmetic for integration into an **NB-IoT chip set** targeting navigation and NTN applications. The receiver achieved **-140 dBm acquisition sensitivity**, -**150 dBm tracking sensitivity**, and a **median positioning accuracy of approximately 7 m**, demonstrating its suitability for deployment in commercial GNSS-enabled applications.

Laboratory Assignments

- Implemented a **fixed-point FIR filter** in **Verilog**.
- Implemented **Newton-Raphson-based division** in **Verilog** using fixed-point arithmetic.
- Simulated **digital modulation schemes** and analysed BER over **AWGN and Rayleigh fading channels** using Python.
- Simulated **MIMO systems** with different **Tx/Rx configurations** and analysed BER performance using Python.
- Implemented **OFDM over AWGN and Rayleigh fading channels** using Python.

RELEVANT COURSES

- **Random Variables and Stochastic Processes :** Statistical modelling, CDF analysis, and receiver sensitivity estimation.
- **Principles of Digital Communications :** BPSK modulation/demodulation, BER analysis, and maximum-likelihood detection.
- **Advanced Digital Signal Processing :** Sampling theory, digital filtering, and up sampling/down sampling techniques.
- **Wireless Communications :** Doppler, carrier phase, and code phase estimation using maximum-likelihood techniques.
- **Matrix Theory :** Nonlinear equation solving and least-squares estimation for GPS PVT computation.

POSITIONS OF RESPONSIBILITY

- **TA for FPGA & DSP lab :** Guided undergrad students in python,verilog and fixed point arthematic.
- **Student Coordinator :** CIRH International Conference.